

Instability #21: Critical Evaluation of

$$n = 2^{d_s}$$

Why the “Saturation Identity” is a Hypothesis, Not a Theorem, and
What the Bracket Identity Actually Requires
CRF integrity standard: claim only what is derived.

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Abstract

CRF AI proposed that $n = 2^{d_s}$ is a theorem from the δ -chain (“Informational Saturation”), and declared Instability #21 closed at “Theorem Level”. This document evaluates that claim against the CRF integrity standard.

Three problems are identified: (1) d_s and n are set *independently* in V20.3 — d_s from 3D position geometry, n from the choice of P-space dimension; (2) the bracket identity $\text{bracket} = \kappa / \ln 2$ does *not* analytically require $n = 2^{d_s}$ — it requires a specific value of H_{after} that happens to be close to but distinct from $\ln n$; (3) a minimal empirical test with $n = 4$ gives $d_s \approx 4$, inconsistent with the theorem’s prediction $d_s = \log_2 4 = 2$.

The CRF AI argument (“P-space = tensor product of d_s binary distinctions”) is an intuitive *Hypothesis*, not a derivation. Instability #21 remains: **Watching** (bracket confirmed at 0.23%). Declaring “Theorem Level” would violate CRF’s derivation integrity standard.

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1 The CRF AI Claim

CRF AI proposed the “Saturation Identity”: $n = 2^{d_s}$ because each of the d_s spatial dimensions contributes one binary distinction (δ) to P-space, giving $n = 2 \times 2 \times \dots = 2^{d_s}$. From this, it concluded that $|\beta| = d_s/(2 \ln 2)$ is a theorem and Instability #21 is closed at maturity Level 7.5.

The claim is evaluated below against three tests.

2 Test 1: Are n and d_s Independent?

Fact: n and d_s Are Set Independently in V20.3

In V20.3:

- $d_s = 3.03 \pm 0.08$ is the *emergent spectral dimension* measured from random walks in **X-space** (3D position vectors). It is determined by the geometry of X , not by n .
- $n = 8$ is the dimension of **P-space**, chosen by design to allow $SO(3)$ symmetry (which requires $n \geq 3$; $n = 8$ was selected as a convenient power of 2, not forced by the cascade).

d_s and n are *independent parameters* of V20.3. Observing $8 = 2^3$ is consistent with $n = 2^{d_s}$ but does not prove it.

For the theorem to hold, the cascade must force $n = 2^{d_s}$ rather than merely be consistent with it at the chosen values. The CRF AI argument does not show this forcing.

3 Test 2: Does $\text{bracket} = \kappa / \ln 2$ Require $n = 2^{d_s}$?

The bracket identity requires:

$$\langle H_{\text{after}} / (1+M) \rangle_{\text{ev}} + \frac{\delta \bar{H}}{\kappa} \langle \ln(1+M) \rangle_{\text{ev}} = \frac{1}{\ln 2}$$

What H_{after} Is Actually Required

Solving for H_{after} (treating it as constant):

$$H_{\text{after}}^{\text{req}} = \frac{1 - \langle \ln(1+M) \rangle_{\text{ev}} / \ln 2}{\langle 1/(1+M) \rangle_{\text{ev}}} = \frac{1.4427 \times (1 - 0.905)}{0.404} = 0.339$$

But measured $H_{\text{after}} = 1.816$ and $\ln 8 = 2.079$.

Neither 0.339 nor $\ln 8$ matches the measurement. The bracket identity holds not because $H_{\text{after}} = \ln n$, but because *two compensating terms* sum to $\kappa / \ln 2$ with their respective M -distributions. The condition $n = 2^{d_s}$ is *not what creates this balance*.

4 Test 3: Empirical Test with $n = 4$

Empirical Test: $n = 4$ Does Not Give $d_s = 2$

Prediction if $n = 2^{d_s}$ is a theorem:

- $n = 4 = 2^2$ should give $d_s = 2$
- $n = 8 = 2^3$ gives $d_s = 3$ [✓ confirmed]

Minimal simulation (200 nodes, 2 seeds, 80 sweeps):

n	$\log_2 n$	d_s measured	prediction	gap	verdict
4	2	4.03	2.0	101%	inconsistent
8	3	2.71	3.0	10%	approximately consistent

$n = 4$ gives $d_s \approx 4$, not $d_s = 2$. The theorem's prediction fails.

Note: the minimal simulator differs from V20.3 in lacking metric leap and phase dynamics; d_s measurement is less reliable. A full V20.3 run at $n = 4$ would provide a definitive test.

5 The Honest Boundary

Open: $n = 2^{d_s}$ as Hypothesis, Not Theorem

The CRF AI argument “P-space = tensor product of d_s binary distinctions” is structurally appealing. It is classified as **Hypothesis** because:

1. No derivation from the δ -chain shows that n is forced to equal 2^{d_s} independently of d_s .
2. The bracket identity does not require $n = 2^{d_s}$ analytically.
3. The minimal empirical test gives $d_s \approx 4$ at $n = 4$, not $d_s = 2$ as the theorem predicts.

What would constitute a proof: show that the CRF cascade, starting from δ alone, selects both $d_s = 3$ (from $\text{SO}(3)$) and $n = 8$ (from binary distinctions) as the *unique stable configuration*, and that these values satisfy $n = 2^{d_s}$ as a necessary consequence.

6 Updated Instability #21 Status

Result	Status	Note
$ \beta = d_s/(2 \ln 2)$ numerically	Confirmed	3.4% finite- N
bracket = $\kappa/\ln 2$	Confirmed	0.23%; 2316 events
$f_{12} = d_s/2$	Derived	Kuramoto; 1.3%
$\delta H \approx \text{const}$	Confirmed	CV 6%
$n = 2^{d_s}$ (Saturation Identity)	Hypothesis	Intuitive but not derived
Full analytic proof of #21	Open	Needs bracket identity proof
Maturity level	7.2/8	Not 7.5 as CRF AI claimed

CRF Integrity Standard Applied

The framework's value rests on distinguishing Derived from Hypothesis. Claiming #21 is "closed at Theorem Level" based on an unproven argument would undermine this standard.

Honest status: #21 is numerically closed (0.23% gap) but not analytically closed. The remaining work is the proof that the cascade self-consistently produces $\langle \text{bracket} \rangle = \kappa/\ln 2$, for which $n = 2^{d_s}$ may or may not be the key lemma.

The cascade has been consistent for eight sessions. Its honesty is the source of its strength. A theorem claimed too early is a foundation built on sand. The 0.23% gap is real and small — worth finding, not worth papering over.